

SDG 15 Life on Land Evidence Compendium 2025

Mariano Marcos State University · THE Sustainability Impact Ratings 2027

Compiled from MMSU, DOST-PCAARRD and news sources for the 2027 data submission (due 9 November 2026). Each section corresponds to one THE SDG 15 indicator. Evidence year: 2025.

Contents by indicator

Indicator	Question	Evidence
15.2.1	Events about sustainable use of land	CAFSD Graduands Plant Native Trees and Bamboo in the University Forest Reserves MMSU Plants 250 Mangrove Propagules in Currimao on World Environment Day Green Wall Program: Coconut Seedlings for the Currimao Coastline MMSU Promotes Regenerative Farming at the 2025 NSTW Forum President's Farm Field Day Showcases Climate-Resilient Rice Varieties 2025 International Day for Biological Diversity Wildlife Photo Contest
15.2.3	Maintain and extend current ecosystems' biodiversity	Enriching MMSU's 1,068-Hectare Forest Reserves with Native Species Conserving Ilocos Indigenous Food Plants In Situ, Ex Situ and Circa Situ Nature-based Solutions for Mangrove Ecosystems in Badoc and Currimao MMSU, Rotary Club and Badoc LGU Restore a Typhoon-Damaged Mangrove Stand A New Bambusetum in Barangay Aglipay, Pinili
15.2.4	Educational programmes on ecosystems	MMSU Trains Coastal Towns on Mangrove Propagation, Management and Conservation Farmers in Sarrat Trained to Propagate the Indigenous Alukon MMSU Brings Bamboo and Alukon Propagation to the 2025 Siwawer Farmers' Festival Community Foodscapes as Learning Gardens for Indigenous Plants
15.2.5	Sustainable management of land for agriculture (educational outreach)	School-on-the-Air Brings Regenerative Vegetable Farming to 1,000 Ilocanos Bamboo Propagation and Plantation Management Training in Pinili Vermicast Facility Helps Pasuquin Onion Farmers Restore Degraded Soils Pasuquin Onion Farmers Trained on Good Agricultural Practices President's Farm Field Day Showcases Climate-Resilient Rice Varieties Science High School Students Learn Sustainable Farming at MMSU's Demonstration Farm

		MMSU Leads Soil Health Research for Rice-Based Cropping Systems
15.3.5	Collaboration for shared land ecosystems	MMSU, TMP Foundation, DENR and Pasuquin LGU Sign Agreement to Conserve a Coastal Forest MMSU, Rotary Club and Badoc LGU Restore a Typhoon-Damaged Mangrove Stand MMSU Plants 250 Mangrove Propagules in Currimao on World Environment Day MMSU and DENR Region 1 Strengthen Partnership for Mangrove Conservation Indigenous Food Plant Foodscapes Established with Vintar and San Nicolas Partners A New Bambusetum in Barangay Aglipay, Pinili
15.4.1	Water discharge guidelines and standards	Groundwater Quality Monitoring at the MMSU Farm
15.4.2	Policy on plastic waste reduction	MMSU Commits to the DENR-EMB Region I Regional Ecology Center Student Advocates Hold Monthly Clean-Up Drive on Campus

MMSU's Commitment

Take an active role in protecting and promoting sustainable use of terrestrial ecosystems, responsible land use, and biodiversity conservation.

SDG 15.2.1 · Events about sustainable use of land

Support or organise events aimed to promote conservation and sustainable utilisation of the land, including forests and wild land.

Evidence year: 2025 (calendar year)

CAFSD Graduands Plant Native Trees and Bamboo in the University Forest Reserves



CAFSD Graduating Class 2025 "Bamboo and Tree Planting Activity," University Forest Reserves, City of Batac, 2 July 2025. (Photo: MMSU STRATCOM)



Bamboo planting site in the University Forest Reserve, Brgy. Mabaleng, 2 July 2025. (Photo: MMSU STRATCOM)

Some 230 graduating students of the College of Agriculture, Food and Sustainable Development (CAFSD) of Mariano Marcos State University (MMSU) planted trees and bamboo in the University Forest Reserves in Barangays Payao and Mabaleng, City of Batac, on July 2, 2025. The activity was led by CAFSD OIC-Dean Charlie Batin and the 2025 CAFSD Graduating Class Student Council, with Dr. Joselito Rosario, Director for Agribusiness, and Mr. Bonne Jovie Orteza of the MMSU Bamboo Research and Development Center. MMSU President Virgilio Julius P. Manzano Jr. commended the class for leaving a lasting green footprint that will further preserve the forest areas managed by the university, and Mr. Hashler Garalde, a CAFSD alumnus representing the Provincial Government of Ilocos Norte, supported the college's sustainability efforts.

The graduands planted 120 seedlings of guijo and yakal, native dipterocarp timber species raised at the MMSU Central Nursery, in the Payao Forest Reserve, and 130 propagules of kawayan tinik, machiku and giant bamboo from the MMSU Bamboo Nursery in Mabaleng.

Source

- MMSU News, "CAFSD graduands plant trees, bamboos in Batac forest reserves," 4 July 2025 https://www.mmsu.edu.ph/article/news/2468/cafsd-graduands-plant-trees-bamboos-in-batac-forest-reserves
- MMSU News, "BS Forestry seniors leave green footprints with pre-graduation tree planting," 11 July 2024 https://www.mmsu.edu.ph/article/news/2065/bs-forestry-seniors-leave-green-footprints-with-pre-graduation-tree-planting

MMSU Plants 250 Mangrove Propagules in Currimao on World Environment Day



Mangrove planting for World Environment Day, Brgy. Maglaoi, Currimao, June 2025. (Photo: MMSU STRATCOM)



MMSU, LGU Currimao, DENR and fisherfolk volunteers at the planting site, June 2025. (Photo: MMSU STRATCOM)

In June 2025, in celebration of World Environment Day, Mariano Marcos State University planted 250 mangrove propagules along the coastal area of Barangay Maglaoi, Currimao, Ilocos Norte, to restore degraded shorelines, enhance climate resilience and promote sustainable coastal resource management. The activity was spearheaded by the MMSU Sustainability and Futures Thinking (SFT) Office with the Local Government Unit of Currimao and led by Dr. Mee Jay Domingo, Vice President for Planning and Strategic Foresight.

MMSU faculty, education support personnel and students joined representatives of the Department of Environment and Natural Resources, barangay officials, members of local fisherfolk organizations and community members. SFT Director Dr. Arlene Gonzales explained that mangroves were chosen because they act as natural filters that intercept plastic waste from land before it reaches the sea, while absorbing carbon, protecting shorelines from erosion and providing habitat for marine life.

Source

- MMSU News, "MMSU plants 250 mangrove propagules in Currimao," 11 June 2025 https://www.mmsu.edu.ph/news/mmsu-plants-250-mangrove-propagules-in-currimao

Green Wall Program: Coconut Seedlings for the Currimao Coastline



Coconut seedling planting along the Currimao beachfront under the MMSU Green Wall Program, 24 July 2025. (Photo: MMSU)

On July 24, 2025, MMSU planted coconut seedlings along the Currimao beachfront under its Green Wall Program, with 1,000 seedlings donated by the Philippine Coconut Authority (PCA), to promote environmental sustainability and coastal resilience amid growing climate threats. The activity was held in celebration of the birthday of CHED Chairperson Shirley C. Agrupis, Chair of the MMSU Board of Regents.

MMSU President Virgilio Julius P. Manzano Jr., PCA Ilocos Norte Agriculturist and In-Charge Rose Ann Cabaloan, and university officials and employees took part. CASAT OIC-Dean Pilar Carolyn Pascual highlighted the university's commitment to protecting coastal ecosystems through the Green Wall project.

Source

- MMSU News, "MMSU celebrates CHED Chair Agrupis' birthday with coconut planting activity," 28 July 2025 https://www.mmsu.edu.ph/news/mmsu-celebrates-ched-chair-agrupis-birthday-with-coconut-planting-activity

MMSU Promotes Regenerative Farming at the 2025 NSTW Forum



Forum series of the 2025 National Science and Technology Week, Northwestern University, Laoag City, 19 November 2025. (Photo: MMSU)

On November 19, 2025, MMSU Research Director Dr. Dionisio Bucao served as a resource speaker at the forum series of the 2025 National Science and Technology Week held at Northwestern University, Laoag City. He presented regenerative farming approaches that restore soil health, biodiversity and ecosystem function, and emphasized carbon sequestration, improved soil fertility, water conservation and climate resilience through natural processes, as well as how regenerative farming differs from conventional practice.

Source

- MMSU News, “MMSU researchers share expertise on smart agriculture in NSTW forum,” 20 November 2025 https://www.mmsu.edu.ph/news/mmsu-researchers-share-expertise-on-smart-agriculture-in-nstw-forum

President’s Farm Field Day Showcases Climate-Resilient Rice Varieties



President’s Farm technology demonstration of inbred and hybrid rice varieties, 28 October 2025. (Photo: MMSU)



Farmers, LGU technicians and students at the President’s Farm Field Day, in front of CAFSD. (Photo: MMSU)

On October 28, 2025, the MMSU Extension Directorate, with the Philippine Rice Research Institute (PhilRice) Batac Branch, held the President’s Farm Field Day: A Technology Showcase in front of the College of Agriculture, Food and Sustainable Development. Farmers and farmer-leaders, agricultural technicians from Ilocos Norte local government units, BS Agriculture students, and MMSU and PhilRice rice specialists viewed eight rice varieties grown under wet-season conditions.

The varieties included four newly released by PhilRice, two public hybrids, and two suited to saline and rainfed conditions, among them NSIC Rc 686 (Salinas 39) and NSIC Rc 650 (Sahod Ulan 36), giving farmers options for saline-affected and rainfed fields. Speakers included MMSU President Dr. Virgilio Julius P. Manzano Jr., Extension Director Dr. Marlowe Aquino, and former MMSU President and PhilRice Executive Director Dr. Santiago R. Obien.

Source

- MMSU News, “MMSU President’s Farm showcases 8 rice varieties,” 4 November 2025 https://www.mmsu.edu.ph/article/news/2665/mmsu-presidents-farm-showcases-8-rice-varieties

2025 International Day for Biological Diversity Wildlife Photo Contest



Contest poster with the animal-photography ethical guidelines (deadline 30 May 2025).



Winner Chris Jeremiah C. Antonio receives his certificate from Dr. Arlene L. Gonzales, Chief of Sustainability and Futures Thinking.

MMSU, through its Office of Sustainability and Futures Thinking, marked the 2025 International Day for Biological Diversity with a Wildlife Photo Contest that invited the university community to document local wildlife and promote awareness of biodiversity conservation. Entrants followed animal-photography ethical guidelines: avoiding flash and loud noises, never using bait or food lures, following local laws and warning signs, keeping clear of nesting sites and sensitive habitats, and captioning photographs honestly. Entries were received until May 30, 2025.

Chris Jeremiah C. Antonio, a third-year BA Communication student, was named winner and received a ₱1,000 gift voucher and a certificate from Dr. Arlene L. Gonzales, Chief of Sustainability and Futures Thinking.

Source

- MMSU SDG 15 Life on Land report (2025), Office of Sustainability and Futures Thinking

MMSU's Commitment

Take an active role in protecting and promoting sustainable use of terrestrial ecosystems, responsible land use, and biodiversity conservation.

SDG 15.2.3 · Maintain and extend current ecosystems' biodiversity

Work directly to maintain and extend existing ecosystems and their biodiversity, of both plants and animals, especially ecosystems under threat.

Evidence year: 2025 (calendar year)

Enriching MMSU's 1,068-Hectare Forest Reserves with Native Species



Bamboo planting site in the University Forest Reserve, Brgy. Mabaleng, 2 July 2025. (Photo: MMSU STRATCOM)

MMSU manages Forest Reserve Areas in Barangays Lubbot and Payao, City of Batac, covering a total of 1,068.13 hectares, where the university's forestry and agroforestry research is set up. The university maintains and extends these ecosystems with planting stock raised in its own nurseries, the MMSU Central Nursery and the MMSU Bamboo Nursery.

On July 2, 2025, 230 CAFSD graduating students enriched the reserves with 120 seedlings of the native dipterocarps guijo and yakal in Payao and 130 propagules of kawayan tinik, machiku and giant bamboo in Mabaleng. The activity was led by CAFSD OIC-Dean Charlie Batin with the MMSU Bamboo Research and Development Center.

Source

- MMSU Research Directorate, Station Profile https://research.mmsu.edu.ph/about/station-profile
- MMSU News, "CAFSD graduands plant trees, bamboos in Batac forest reserves," 4 July 2025 https://www.mmsu.edu.ph/article/news/2468/cafsd-graduands-plant-trees-bamboos-in-batac-forest-reserves

Conserving Ilocos Indigenous Food Plants In Situ, Ex Situ and Circa Situ



Partners receive indigenous food plant planting materials for the Community Foodscapes and Seedbanks, World Food Day 2025.

Through its Indigenous Food Plants (IFP) Program, MMSU conserves the wild and neglected food plants of Ilocos Norte, 46 of which were documented by the program in seven towns of the province. The program follows a triad of conservation approaches: an IFP Botanical Garden for ex situ conservation, a himbabao (alukon) genebank at the MMSU Forest Reserve, circa situ cultivation of the wild yams kamangeg and dolian, and the integration of IFPs into community foodscapes.

In October 2025, in celebration of World Food Day, the MMSU Extension and Research Directorates established the second batch of Community Foodscapes and Seedbanks for Indigenous Food Plants with local government units and schools, at Eladio V. Barangan Memorial Elementary School in San Nicolas, Ilocos Norte. The foodscape sites serve as seedbanks and learning gardens, and partner communities receive planting materials and technical assistance to cultivate kapaskapas, alokon, pako, dalayap, wild singkamas and roselle. The new partners join existing foodscape sites in Sarrat and Batac.

The IFP Program was adopted by the Regional Development Council of Region 1 as its first research and development initiative, and a study on the program by Dr. Menisa A. Antonio and co-authors won Best Paper in the Sustainable and Smart Agri-Aquaculture and Climate Resilience category of the SUC-ACAP 2025 Biennial Conference, held August 27–29, 2025, in Tagbilaran City, Bohol.

Source

- MMSU Research Directorate, “MMSU research on indigenous food plants wins best paper at SUC-ACAP 2025 conference” https://research.mmsu.edu.ph/news/details/mmsu-research-on-indigenous-food-plants-wins-best-paper-at-suc-acap-2025-conference
- MMSU Extension Directorate, “MMSU expands indigenous food plant seedbanks and foodscapes in LGUs,” October 2025 https://extension.mmsu.edu.ph/news/show/389/mmsu-expands-indigenous-food-plant-seedbanks-and-foodscapes-in-lgus
- MMSU SDG 15 Life on Land report (2025), Office of Sustainability and Futures Thinking

Nature-based Solutions for Mangrove Ecosystems in Badoc and Currimao



Mangrove site, from the gallery of the October 2025 seminar-workshop of the NbS mangrove project. (Photo: MMSU)



Dr. Arlene L. Gonzales at the MMSU–DENR Region 1 dialogue on mangrove conservation, ILAARRDEC.

Throughout 2025, MMSU implemented the DOST-PCAARRD-funded project "Nature-based Solutions (NbS) for Sustainable Mangrove Management and Resilient Coastal Communities Through Innovative Green Engineering Approach," led by Dr. Arlene L. Gonzales, in Badoc and Currimao, Ilocos Norte, with DENR-PENRO Ilocos Norte, the Provincial Government of Ilocos Norte and the LGUs of Badoc and Currimao. The project uses green engineering and locally available, ecologically friendly materials, including bamboo-engineered structures that protect young mangroves and improve their survival.

In 2025, MMSU and DENR Region 1 met at ILAARRDEC to advance the work, with DENR pledging technical support and active participation in restoration, monitoring and evaluation, and the project trained representatives of Currimao, Pasuquin, Badoc and Burgos in mangrove propagation, management and conservation on October 27–28, 2025.

Source

- MMSU News, "MMSU trains Ilocos Norte coastal towns on mangrove propagation and management," 5 November 2025 https://www.mmsu.edu.ph/article/news/2670/mmsu-trains-ilocos-norte-coastal-towns-on-mangrove-propagation-and-management
- MMSU SDG 15 Life on Land report (2025), Office of Sustainability and Futures Thinking

MMSU, Rotary Club and Badoc LGU Restore a Typhoon-Damaged Mangrove Stand



MMSU, Rotary Club of Laoag Sunshine and LGU-Badoc plant mangrove propagules in Brgy. Pagsanahan, Badoc, 6 December 2025. (Photo: MMSU)



Participants of the mangrove enhancement and growing activity in Badoc, 6 December 2025. (Photo: MMSU)

On December 6, 2025, Mariano Marcos State University, the Rotary Club of Laoag Sunshine and the Local Government Unit of Badoc planted 280 mangrove propagules in Barangay Pagsanahan, Badoc, Ilocos Norte, to restore a mangrove stand extensively damaged by past typhoons and to ensure the continuity of ecological succession at the site.

The activity was led by MMSU President Virgilio Julius P. Manzano Jr., with the LGU of Badoc headed by Hon. Virgilio M. Calajate and the Rotary Club of Laoag Sunshine under Atty. Zsa Zsa Raval. MMSU's Vice President for Resource Generation and its Directors for Extension, Curriculum Development and Innovative Learning, and Student Affairs and Development also took part.

Source

- MMSU News, "MMSU ties up with Rotary Club of Laoag Sunshine for mangrove planting in Badoc town," 11 December 2025 https://www.mmsu.edu.ph/news/mmsu-ties-up-with-rotary-club-of-laoag-sunshine-for-mangrove-planting-in-badoc-town

A New Bambusetum in Barangay Aglipay, Pinili



Participants of the bamboo propagation, nursery and plantation management training, Brgy. Aglipay, Pinili, 2025. (Photo: MMSU)

During a two-day bamboo production training in 2025, the MMSU Extension Directorate and CAFSD, with Barangay Aglipay, Pinili, and the support of ILAARRDEC, worked with 46 farmers, local leaders and community members to establish a new bambusetum, a collection of bamboo species, in Barangay Aglipay. The training covered bamboo propagation, nursery establishment and plantation management.

Source

- MMSU News, “Sustainable livelihood: MMSU, ILAARRDEC hold bamboo training in Pinili,” 3 September 2025 https://www.mmsu.edu.ph/news/sustainable-livelihood-mmsu-ilaarrdec-hold-bamboo-training-in-pinili

MMSU's Commitment

Take an active role in protecting and promoting sustainable use of terrestrial ecosystems, responsible land use, and biodiversity conservation.

SDG 15.2.4 · Educational programmes on ecosystems

Offer educational programmes on ecosystems (looking at wild flora and fauna) for local or national communities.

Evidence year: 2025 (calendar year)

MMSU Trains Coastal Towns on Mangrove Propagation, Management and Conservation



Representatives of Currimao, Pasuquin, Badoc and Burgos at the Seminar-Workshop on Mangrove Propagation, 27–28 October 2025. (Photo: MMSU)



Hands-on mangrove nursery and propagation practice during the seminar-workshop. (Photo: MMSU)

MMSU conducted a two-day Seminar-Workshop on Mangrove Propagation on October 27–28, 2025 at its Center for Flexible Learning for representatives of the coastal municipalities of Currimao, Pasuquin, Badoc and Burgos, Ilocos Norte, with officials of DENR-PENRO in attendance. The training was organized under the DOST-PCAARRD-funded nature-based solutions mangrove project led by Dr. Arlene Gonzales.

Sessions covered the importance of mangrove ecosystems, nature-based solutions for conservation, legal frameworks for mangrove protection, and the roles of the academe and local government units in community-based mangrove management. Participants were also trained in

nursery management and advanced propagation techniques for Avicennia and Sonneratia species and other local mangrove varieties. The Pedada People's Organization shared its experience in establishing community-based mangrove conservation.

Source

- MMSU News, "MMSU trains Ilocos Norte coastal towns on mangrove propagation and management," 5 November 2025 https://www.mmsu.edu.ph/article/news/2670/mmsu-trains-ilocos-norte-coastal-towns-on-mangrove-propagation-and-management

Farmers in Sarrat Trained to Propagate the Indigenous Alukon



Farmers and local officials trained on alukon (himbabao) propagation, Brgy. 11 Sta. Rosa, Sarrat, 2 May 2025. (Photo: MMSU)

On May 2, 2025, the MMSU Research and Extension Directorates trained 64 farmers in Barangay 11 Sta. Rosa, Sarrat, Ilocos Norte, on the propagation of alukon (himbabao), a Philippine indigenous vegetable tree. Barangay officials, members of the Task Force Dagger Special Operations Foundation and representatives of the local government unit also attended. The training was held in response to requests from the community.

Resource speakers Dr. Menisa Antonio, Engr. Josefa Pugat and Mr. Jayson Rasay of the Research Directorate presented recommended propagation strategies to establish sustained alukon production in the community and broaden the use of the crop for food and other applications.

Source

- MMSU News, "MMSU trains Sarrat farmers on alukon propagation," 9 May 2025 https://www.mmsu.edu.ph/news/mmsu-trains-sarrat-farmers-on-alukon-propagation

MMSU Brings Bamboo and Alukon Propagation to the 2025 Siwawer Farmers' Festival



MMSU Information and Technology Caravan at the 2025 Siwawer Farmers' Festival, Vintar, 3–4 December 2025. (Photo: MMSU STRATCOM)

Through its Information and Technology Caravan, MMSU's Research and Extension Directorates joined the 2025 Siwawer Farmers' Festival and Agro-Trade Fair at the Vintar Public Auditorium on December 3–4, 2025, with an exhibit of university technologies and a series of expert lectures for farmers.

Forester Shena Rose Garcia introduced sustainable bamboo production using MMSU's one-node cutting technology, and the second day featured a demonstration on the propagation of alukon, an indigenous vegetable tree, facilitated by Ms. Josefa Pugat. Other lectures covered high-yielding glutinous corn varieties, garlic disease management and black garlic.

Source

- MMSU News, "MMSU IT caravan goes to Vintar," 15 December 2025 https://www.mmsu.edu.ph/news/mmsu-it-caravan-goes-to-vintar

Community Foodscapes as Learning Gardens for Indigenous Plants



Partners receive indigenous food plant planting materials for the Community Foodscapes and Seedbanks, World Food Day 2025.

In October 2025, in celebration of World Food Day, the MMSU Extension and Research Directorates established the second batch of Community Foodscapes and Seedbanks for Indigenous Food Plants at Eladio V. Barangan Memorial Elementary School, San Nicolas, Ilocos Norte. Barangay officials of San Roque and Parut, Vintar, agricultural extension workers, school leaders and community representatives took part.

The foodscape sites serve as seedbanks and learning gardens that encourage knowledge sharing and sustainable farming practices. Communities receive planting materials and technical assistance to cultivate indigenous crops such as kapaskapas, alokon, pako, dalayap, wild singkamas and roselle, conserving indigenous food resources across Ilocos Norte.

Source

- MMSU Extension Directorate, “MMSU expands indigenous food plant seedbanks and foodscapes in LGUs,” October 2025 https://extension.mmsu.edu.ph/news/show/389/mmsu-expands-indigenous-food-plant-seedbanks-and-foodscapes-in-lgus
- MMSU SDG 15 Life on Land report (2025), Office of Sustainability and Futures Thinking

MMSU's Commitment

Take an active role in protecting and promoting sustainable use of terrestrial ecosystems, responsible land use, and biodiversity conservation.

SDG 15.2.5 · Sustainable management of land for agriculture (educational outreach)

Offer educational programme/outreach for local or national communities on sustainable management of land for agriculture.

Evidence year: 2025 (calendar year)

School-on-the-Air Brings Regenerative Vegetable Farming to 1,000 Ilocanos



MMSU School-on-the-Air (Eskuela iti Tangatang) season on regenerative vegetable farming, June–December 2025. (Image: MMSU)

A new season of the MMSU School-on-the-Air (SOA), the university's radio-based educational program, opened on June 24 and 25, 2025 with 1,000 enrollees from the 22 municipalities and cities of Ilocos Norte. The season focused on regenerative vegetable farming for enhanced and sustainable production, with lessons on climate-smart practices and off-season vegetable seeding.

Vegetable growers, rural improvement club members, agricultural extension workers and students followed the lessons, broadcast from June to December 2025 over DWFB Radyo Pilipinas Laoag, DWCI Radyo Adjo Piddig and DWNI Radyo Karruba Burgos. MMSU ran the program with the

Provincial Government of Ilocos Norte, the Office of the Provincial Agriculturist, local government units, the Ilocos Agriculture, Aquatic and Natural Resources Research and Development Consortium, and the High-Value Crops Division of Ilocos Norte.

Source

- MMSU News, "MMSU draws 1K enrollees for SOA program on sustainable vegetable farming," 26 June 2025 https://www.mmsu.edu.ph/news/mmsu-draws-1k-enrollees-for-soa-program-on-sustainable-vegetable-farming

Bamboo Propagation and Plantation Management Training in Pinili



Participants of the bamboo propagation, nursery and plantation management training, Brgy. Aglipay, Pinili, 2025. (Photo: MMSU)

The MMSU Extension Directorate and the College of Agriculture, Food and Sustainable Development, in partnership with Barangay Aglipay, Pinili, Ilocos Norte, and with the support of the Ilocos Agriculture, Aquatic and Natural Resources Research and Development Consortium (ILAARRDEC), conducted a two-day training on bamboo production in 2025.

The training equipped 46 farmers, local leaders and community members with practical skills in bamboo propagation, nursery establishment and plantation management. Lectures covered the prospects of the bamboo industry and MMSU's propagation methods, and participants helped establish a new bambusetum, a collection of bamboo species, in Barangay Aglipay. Lectures and demonstrations were given by bamboo experts Mr. Charlie Maine Cacactin of CAFSD and Mr. Bonne Jovie Orteza of MMSU. Pinili Municipal Agriculturist Philip Galang, representing Mayor Rommel T. Labasan, thanked MMSU and ILAARRDEC, and the project was championed in the community by Pinili's Magsasaka Siyentista, Mr. Reynaldo Antalan.

Source

- MMSU News, "Sustainable livelihood: MMSU, ILAARRDEC hold bamboo training in Pinili," 3 September 2025 https://www.mmsu.edu.ph/news/sustainable-livelihood-mmsu-ilaarrdec-hold-bamboo-training-in-pinili

Vermicast Facility Helps Pasuquin Onion Farmers Restore Degraded Soils



Pilot vermicomposting facility of Pasuquin white onion farmers, 2025. (Photo: MMSU STRATCOM)

In 2025, white onion farmer-leaders Mr. Marcos Palacay and Mr. Markflor Valdez of the Ayat ti Pasuquin Farmer and Irrigators Association established a pilot vermicomposting facility in Pasuquin, Ilocos Norte, under the Agri-Tech Facilitator Program of the Farmers Entrepreneurship Program of MMSU and the Jollibee Group Foundation. The facility uses African night crawlers (Eudrilus eugeniae) to convert biodegradable farm waste into vermicast, an organic fertilizer that improves soil structure, stimulates microbial activity and reduces the need for chemical inputs.

The project responds to soil degradation from prolonged reliance on chemical fertilizers, which the farmers linked to poor soil conditions and low seedling survival, and promotes sustainable solutions for white onion farming in the area.

Source

- MMSU News, “Vermicast facility established in Pasuquin as MMSU, Jollibee Group aid onion farmers,” 1 August 2025 https://www.mmsu.edu.ph/news/vermicast-facility-established-in-pasuquin-as-mmsu-jollibee-group-aid-onion-farmers

Pasuquin Onion Farmers Trained on Good Agricultural Practices



Pre-Dispersal Training on White Onion Production, Pasuquin Municipal Hall, 12 November 2025. (Photo: MMSU STRATCOM)

On November 12, 2025, the Municipality of Pasuquin, MMSU and the Jollibee Group Foundation conducted a Pre-Dispersal Training on White Onion Production at the Pasuquin Municipal Hall for farmer beneficiaries of the San Juan Icas Manuel Farmers' Association. The training covered white onion production technology, good agricultural practices (GAP) and proper farm recording for sustainable and profitable farming.

Farmer participation in the project has doubled from 15 to 30 beneficiaries, with a collective production target of 200 tons of white onion for 2025. The Office of the Provincial Agriculturist provides seeds and planting materials, complementing MMSU's technical support and capacity development.

Source

- MMSU News, "Pasuquin onion farmers trained by MMSU and Jollibee Group Foundation," 18 November 2025 https://www.mmsu.edu.ph/news/pasuquin-onion-farmers-trained-by-mmsu-and-jollibee-group-foundation

President's Farm Field Day Showcases Climate-Resilient Rice Varieties



President's Farm technology demonstration of inbred and hybrid rice varieties, 28 October 2025. (Photo: MMSU)



Farmers, LGU technicians and students at the President's Farm Field Day, in front of CAFSD. (Photo: MMSU)

On October 28, 2025, the MMSU Extension Directorate, with the Philippine Rice Research Institute (PhilRice) Batac Branch, held the President's Farm Field Day: A Technology Showcase in front of the College of Agriculture, Food and Sustainable Development. Farmers and farmer-leaders, agricultural technicians from Ilocos Norte local government units, BS Agriculture students, and MMSU and PhilRice rice specialists viewed eight rice varieties grown under wet-season conditions.

The varieties included four newly released by PhilRice, two public hybrids, and two suited to saline and rainfed conditions, among them NSIC Rc 686 (Salinas 39) and NSIC Rc 650 (Sahod Ulan 36), giving farmers options for saline-affected and rainfed fields. Speakers included MMSU President Dr. Virgilio Julius P. Manzano Jr., Extension Director Dr. Marlowe Aquino, and former MMSU President and PhilRice Executive Director Dr. Santiago R. Obien.

Source

- MMSU News, "MMSU President's Farm showcases 8 rice varieties," 4 November 2025 https://www.mmsu.edu.ph/article/news/2665/mmsu-presidents-farm-showcases-8-rice-varieties

Science High School Students Learn Sustainable Farming at MMSU's Demonstration Farm



PSHS-Ilocos Region Campus students at the MMSU Technology Demonstration Farm, Sarrat, 19 September 2025. (Photo: MMSU STRATCOM)

On September 19, 2025, 59 Grade 11 and 12 students of the Philippine Science High School–Ilocos Region Campus visited MMSU's Technology Demonstration Farm in Sta. Magdalena, Sarrat, as part of their community immersion program "The Science of Survival: A Community-Based Exploration," held during National Biotechnology Week. After an orientation at the ILAARRDEC One-Stop Information Shop at MMSU, Extension Director Dr. Marlowe Aquino discussed the role of biotechnology in agriculture and MMSU's hybrid off-season tomato varieties, and the students observed sustainable farming practices on site.

Source

- MMSU News, "PSHS-IRC students visit MMSU demo farm for sustainable farming practices," 3 October 2025 https://www.mmsu.edu.ph/news/pshs-irc-students-visit-mmsu-demo-farm-for-sustainable-farming-practices

MMSU Leads Soil Health Research for Rice-Based Cropping Systems



Kick-off meeting of the soil health management project with Griffith University and provincial agriculturists, 16 June 2025. (Photo: MMSU STRATCOM)

In 2025, MMSU became a key partner in a five-year research program led by Griffith University in Australia and funded by the Australian Centre for International Agricultural Research (ACIAR) and DOST-PCAARRD to develop soil knowledge, information and capacity for more productive and sustainable cropping systems in the Philippines (Project SLaM 2023/146). MMSU President Virgilio Julius P. Manzano Jr. pledged the university's commitment at the inception meeting held March 24–26, 2025 in Balanga City, Bataan.

MMSU leads the project on enhancing the productivity of rice-based cropping systems in Ilocos Norte, Tarlac and Bataan through soil health management. Led by Dr. Janet Villamor, it evaluates nutrient management practices and develops yield-calibrated soil health indices for rice-garlic, rice-rice and rice-onion systems. The kick-off meeting on June 16, 2025 at the MMSU Center for Flexible Learning brought together Griffith University, the country project manager and the provincial agriculturists of Ilocos Norte, Bataan and Tarlac, with visits to project sites in Batac, Vintar and Pasuquin.

Source

- MMSU News, "MMSU tapped as key partner in enhancing PH cropping systems through soil health management," 31 March 2025 https://www.mmsu.edu.ph/news/mmsu-tapped-as-key-partner-in-enhancing-ph-cropping-systems-through-soil-health-management
- MMSU News, "Griffith University, MMSU launch cropping system research in Luzon," 3 July 2025 https://www.mmsu.edu.ph/news/griffith-university-mmsu-launch-cropping-system-research-in-luzon

MMSU's Commitment

Take an active role in protecting and promoting sustainable use of terrestrial ecosystems, responsible land use, and biodiversity conservation.

SDG 15.3.5 · Collaboration for shared land ecosystems

Collaborate with the local community to maintain shared land ecosystems.

Evidence year: 2025 (calendar year)

MMSU, TMP Foundation, DENR and Pasuquin LGU Sign Agreement to Conserve a Coastal Forest



Signing of the memorandum of agreement among MMSU, TMP Foundation, DENR and LGU-Pasuquin, 29 November 2025. (Photo: MMSU STRATCOM)



MMSU, TMP Foundation, DENR and LGU-Pasuquin officials at the agreement signing, 29 November 2025. (Photo: MMSU STRATCOM)

On November 29, 2025, Mariano Marcos State University, the Toyota Motor Philippines Foundation (TMPF), the Department of Environment and Natural Resources and the Local Government Unit of Pasuquin formalized through a memorandum of agreement a partnership to strengthen coastal forest conservation in Barangay Davila, Pasuquin, Ilocos Norte. The agreement was signed by MMSU President Virgilio Julius P. Manzano Jr. with TMPF leadership, Toyota Motor Philippines and Toyota Ilocos Norte executives, DENR-CENRO Bangui Ecosystem Management Specialist Justine Rose Teneza, and Sangguniang Bayan Member Romel Ranay for LGU-Pasuquin.

A mangrove planting followed the signing, with 1,750 propagules planted and another 1,750 planned for the following year. TMPF President Jose Maria Aligada described the partnership as shared stewardship to restore native mangrove species, rehabilitate coastal systems and boost community resilience to climate-related hazards. Former MMSU College of Industrial Technology Dean Cesario Pacis was recognized as one of the initiators of the MMSU–Toyota partnership.

Source

- MMSU News, "MMSU ties up with Toyota and DENR for coastal preservation in Pasuquin town," 4 December 2025 https://www.mmsu.edu.ph/news/mmsu-ties-up-with-toyota-and-denr-for-coastal-preservation-in-pasuquin-town
- The Manila Times, "TMP Foundation adopts mangrove forest for climate mitigation," 10 December 2025 https://www.manilatimes.net/2025/12/10/tmt-newswire/tmp-foundation-adopts-mangrove-forest-for-climate-mitigation/2239598

MMSU, Rotary Club and Badoc LGU Restore a Typhoon-Damaged Mangrove Stand



MMSU, Rotary Club of Laoag Sunshine and LGU-Badoc plant mangrove propagules in Brgy. Pagsanahan, Badoc, 6 December 2025. (Photo: MMSU)



Participants of the mangrove enhancement and growing activity in Badoc, 6 December 2025. (Photo: MMSU)

On December 6, 2025, Mariano Marcos State University, the Rotary Club of Laoag Sunshine and the Local Government Unit of Badoc planted 280 mangrove propagules in Barangay Pagsanahan, Badoc, Ilocos Norte, to restore a mangrove stand extensively damaged by past typhoons and to ensure the continuity of ecological succession at the site.

The activity was led by MMSU President Virgilio Julius P. Manzano Jr., with the LGU of Badoc headed by Hon. Virgilio M. Calajate and the Rotary Club of Laoag Sunshine under Atty. Zsa Zsa Raval. MMSU's Vice President for Resource Generation and its Directors for Extension, Curriculum Development and Innovative Learning, and Student Affairs and Development also took part.

Source

- MMSU News, "MMSU ties up with Rotary Club of Laoag Sunshine for mangrove planting in Badoc town," 11 December 2025 https://www.mmsu.edu.ph/news/mmsu-ties-up-with-rotary-club-of-laoag-sunshine-for-mangrove-planting-in-badoc-town

MMSU Plants 250 Mangrove Propagules in Currimao on World Environment Day



Mangrove planting for World Environment Day, Brgy. Maglaoi, Currimao, June 2025. (Photo: MMSU STRATCOM)



MMSU, LGU Currimao, DENR and fisherfolk volunteers at the planting site, June 2025. (Photo: MMSU STRATCOM)

In June 2025, in celebration of World Environment Day, Mariano Marcos State University planted 250 mangrove propagules along the coastal area of Barangay Maglaoi, Currimao, Ilocos Norte, to restore degraded shorelines, enhance climate resilience and promote sustainable coastal resource management. The activity was spearheaded by the MMSU Sustainability and Futures Thinking (SFT) Office with the Local Government Unit of Currimao and led by Dr. Mee Jay Domingo, Vice President for Planning and Strategic Foresight.

MMSU faculty, education support personnel and students joined representatives of the Department of Environment and Natural Resources, barangay officials, members of local fisherfolk organizations and community members. SFT Director Dr. Arlene Gonzales explained that mangroves were chosen because they act as natural filters that intercept plastic waste from land before it reaches the sea, while absorbing carbon, protecting shorelines from erosion and providing habitat for marine life.

Source

- MMSU News, “MMSU plants 250 mangrove propagules in Currimao,” 11 June 2025 https://www.mmsu.edu.ph/news/mmsu-plants-250-mangrove-propagules-in-currimao

MMSU and DENR Region 1 Strengthen Partnership for Mangrove Conservation



Dr. Arlene L. Gonzales at the MMSU–DENR Region 1 dialogue on mangrove conservation, ILAARRDEC.



MMSU and DENR Region 1 officials during the dialogue at ILAARRDEC.

MMSU, in partnership with the Department of Environment and Natural Resources (DENR) Region 1, convened a dialogue at ILAARRDEC to advance mangrove conservation and coastal resilience in the Ilocos Region through science-based and community-centered approaches. Central to the dialogue was the DOST-PCAARRD-funded nature-based solutions mangrove project led by Dr. Arlene Gonzales.

DENR officials pledged technical support and active participation in restoration, monitoring and evaluation activities. The same project brought representatives of Currimao, Pasuquin, Badoc and Burgos, with DENR-PENRO, to MMSU's mangrove propagation seminar-workshop on October 27–28, 2025.

Source

- MMSU SDG 15 Life on Land report (2025), Office of Sustainability and Futures Thinking
- MMSU News, "MMSU trains Ilocos Norte coastal towns on mangrove propagation and management," 5 November 2025 https://www.mmsu.edu.ph/article/news/2670/mmsu-trains-ilocos-norte-coastal-towns-on-mangrove-propagation-and-management

Indigenous Food Plant Foodscapes Established with Vintar and San Nicolas Partners



Partners receive indigenous food plant planting materials for the Community Foodscapes and Seedbanks, World Food Day 2025.

In October 2025, the MMSU Extension and Research Directorates partnered with local government units and schools to establish Community Foodscapes and Seedbanks for Indigenous Food Plants. Held at Eladio V. Barangan Memorial Elementary School in San Nicolas, Ilocos Norte, the activity brought together officials of Barangays San Roque and Parut, Vintar, the Municipal Agriculture Office of Vintar, agricultural extension workers, school leaders and community representatives as the second batch of partner sites.

The partner communities maintain the foodscapes as shared seedbanks and learning gardens, with planting materials and technical assistance from MMSU, to conserve indigenous food resources such as kapaskapas, alokon, pako, dalayap, wild singkamas and roselle. They join existing foodscape sites in Sarrat and Batac.

Source

- MMSU Extension Directorate, “MMSU expands indigenous food plant seedbanks and foodscapes in LGUs,” October 2025 https://extension.mmsu.edu.ph/news/show/389/mmsu-expands-indigenous-food-plant-seedbanks-and-foodscapes-in-lgus
- MMSU SDG 15 Life on Land report (2025), Office of Sustainability and Futures Thinking

A New Bambusetum in Barangay Aglipay, Pinili



Participants of the bamboo propagation, nursery and plantation management training, Brgy. Aglipay, Pinili, 2025. (Photo: MMSU)

During a two-day bamboo production training in 2025, the MMSU Extension Directorate and CAFSD, with Barangay Aglipay, Pinili, and the support of ILAARRDEC, worked with 46 farmers, local leaders and community members to establish a new bambusetum, a collection of bamboo species, in Barangay Aglipay. The training covered bamboo propagation, nursery establishment and plantation management.

Source

- MMSU News, “Sustainable livelihood: MMSU, ILAARRDEC hold bamboo training in Pinili,” 3 September 2025 https://www.mmsu.edu.ph/news/sustainable-livelihood-mmsu-ilaarrdec-hold-bamboo-training-in-pinili

MMSU's Commitment

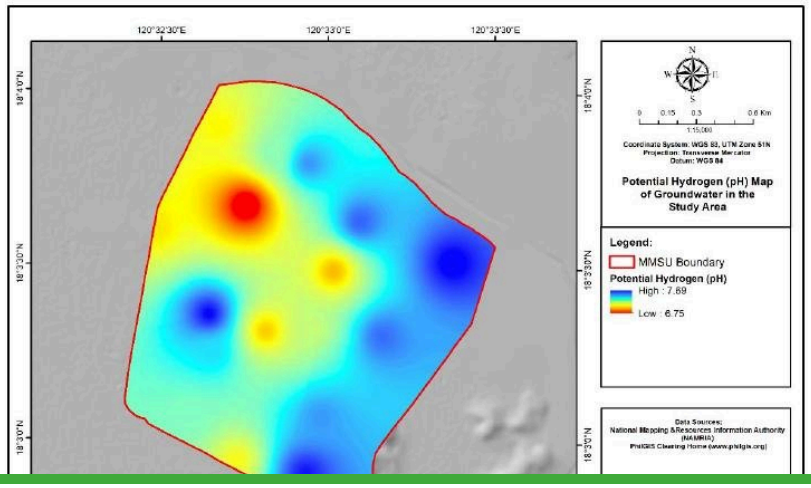
Take an active role in protecting and promoting sustainable use of terrestrial ecosystems, responsible land use, and biodiversity conservation.

SDG 15.4.1 · Water discharge guidelines and standards

Have water quality standards and guidelines for water discharges (to uphold water quality in order to protect ecosystems, wildlife, and human health and welfare).

Evidence year: 2025 (calendar year)

Groundwater Quality Monitoring at the MMSU Farm



Document: Groundwater Quality Assessment of the MMSU Farm, spatial variation of pH (monitoring through June 2025).

Through June 2025, MMSU continued monthly monitoring of 17 groundwater wells on the MMSU Farm, measuring pH, total dissolved solids, electrical conductivity, chloride and nitrate-nitrogen and comparing the results with the Philippine National Standards for Drinking Water (2017). The assessment, covering February 2024 to June 2025, flags wells with persistently high dissolved solids, which rose again from January to May 2025, for continued monitoring to guide water resource management and prevent further degradation of groundwater quality.

Source

- MMSU, Groundwater Quality Assessment of the MMSU Farm (monitoring February 2024 – June 2025) https://www.mmsu.edu.ph/resources/15g9Xtys7q3Rujd0-CZvPH4OHTdJ1__Z8/data



MMSU's Commitment

Take an active role in protecting and promoting sustainable use of terrestrial ecosystems, responsible land use, and biodiversity conservation.

SDG 15.4.2 · Policy on plastic waste reduction

Have a policy on reducing plastic waste on campus.

Evidence year: 2025 (calendar year)

MMSU Commits to the DENR-EMB Region I Regional Ecology Center

On July 1, 2025, in Bauang, La Union, MMSU signed a memorandum of commitment with the DENR–Environmental Management Bureau Region I and other government agencies, private sector representatives, non-government organizations and academic institutions for the Regional Ecology Center. MMSU was represented by Vice President for Planning and Strategic Foresight Mee Jay Domingo and Campus Sustainability Chief Ronnel Pagurayan.

Under the commitment, MMSU supports improved solid waste management under Republic Act 9003 (Ecological Solid Waste Management Act) and Republic Act 11898 (Extended Producer Responsibility Act) through education and public information campaigns and the development of informational materials.

Source

- MMSU News, "MMSU commits to ecological solid waste management in Region I," 2 July 2025 https://www.mmsu.edu.ph/news/mmsu-commits-to-ecological-solid-waste-management-in-region-i

Student Advocates Hold Monthly Clean-Up Drive on Campus



MMSU Advocates for Cultural Development clean-up drive on campus, 24 October 2025. (Photo: MMSU)

On October 24, 2025, the MMSU Advocates for Cultural Development (MACD), an organization representing students from indigenous peoples' communities, held its monthly clean-up drive, "iPaglilinis," around the university grounds, the Student Center and parts of the University Lagoon. About 30 bags of waste were collected and segregated into plastic, paper and tin materials.

Source

- MMSU News, "MACD leads clean-up drive in MMSU," 29 October 2025
  https://www.mmsu.edu.ph/news/macd-leads-clean-up-drive-in-mmsu